

Multiple Choice

1. **Answer:** E

Options: A) $t^2 e^t$; B) $t e^2 t$; C) $e(-t)$; D) $t^3 e^t$; E) te^t

Note: Correct option: E - te^t

2. **Answer:** A

Options: A) (40.74, -20.17, 12); B) (- 30.74, 10.17, - 2); C) (0, 40.34, 24); D) (30.74, 10.17, 2); E) (40.74, 20.17, -12)

Note: Correct option: A - (40.74, -20.17, 12)

3. **Answer:** A

Options: A) $1000 \sin 10^6 t$; B) $100 \sin 10^6 t$; C) $50 \cos 10^6 t$; D) $1000 \cos 10^6 t$; E) $0.1 \cos 10^6 t$

Note: Correct option: A - $1000 \sin 10^6 t$

4. **Answer:** D

Options: A) $(1 - e^{(-2 t)}) / t$; B) $(2 \sinh t) / t^2$; C) $(2 \tanh t) / t$; D) $(2 \sinh t) / t$; E) $(2 \tanh t) * t$

Note: Correct option: D - $(2 \sinh t) / t$

5. **Answer:** E

Options: A) 150; B) 100; C) 200; D) 5; E) 10

Note: Correct option: E - 10

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '74.6%'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'XOR, XNOR, NOR.'

Long Form Response

11. **Answer:** 239 microfarads. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 172.60 ft^3 . Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key